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Transforming Historical Drilling Data into Strategic Insights with Large Language Models

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Abstract

This paper explores the integration of Large Language Models (LLMs), such as DeepSeek, with unstructured data repositories in upstream petroleum operations. The goal is to develop an LLM-driven assistant that processes historical drilling documents to support strategic well planning, with a focus on risk mitigation and operational efficiency.

Our approach involves combining a Retrieval-Augmented Generation (RAG) pipeline with an LLM to interpret unstructured daily drilling reports. Embedding of historical content is performed using a domain-optimized model (BGE-Mining/v1) and stored in a vector database for semantic retrieval. The assistant retrieves relevant historical narratives based on queries about specific drilling challenges or spatial intervals (e.g., nearby wells or formations). Upon receiving a user query, the system returns structured insights such as prior incidents, intervals of concern, and recommended corrective measures.

Preliminary tests using a carefully designed corpus demonstrated that the LLM-based assistant can reliably identify recurring drilling risks, such as stuck pipe events, abnormal flow signatures, pressure-related anomalies, and improper mud weight reductions. These insights were extracted from simulated field data constructed to replicate authentic operational reporting styles, including realistic technical language, domain-specific abbreviations, and event sequences. The system effectively interpreted both explicit incidents and subtle early-warning signs, such as pit gain trends and SICP fluctuations. Responses to user queries were generated within minutes and achieved high scores in expert evaluations for contextual accuracy and technical clarity. Furthermore, the architecture incorporated spatial filtering mechanisms, enabling the assistant to simulate retrieval of offset well data based on user-defined coordinates or formation intervals. This spatial logic was essential for testing the assistant's ability to prioritize relevant records within a given geographical scope, further validating its application to location-aware risk assessment tasks in pre-drill planning.

This framework highlights the potential of LLMs to transform pre-drill planning workflows by surfacing actionable lessons from complex, text-heavy drilling archives. The proposed system demonstrates a novel and practical application of AI in upstream planning, enabling engineers to proactively manage risks using historical knowledge through a natural language interface. Future enhancements may further

increase accuracy and reliability by incorporating modular components that independently handle retrieval, validation, and reasoning.

Introduction

In upstream oil and gas operations, the success of a drilling campaign depends heavily on the ability to identify and mitigate geological and operational risks during the planning phase. Historical drilling reports contain valuable insights into formation behavior, well control events, and prior mitigation strategies—but much of this information remains locked in free-text narratives spread across thousands of documents. Traditional keyword-based search tools often fall short when engineers need contextual answers tied to specific formations or geographic locations.

The primary objective of this study is to evaluate whether Large Language Models (LLMs), when combined with Retrieval-Augmented Generation (RAG) techniques, can significantly enhance the extraction of actionable insights from historical drilling documentation, thereby improving pre-drill planning efficiency and risk mitigation in complex geological settings.

Specifically, this research tests the following hypotheses:

1. An LLM-powered assistant can substantially reduce the time required to retrieve relevant historical drilling information compared to traditional keyword-based search and manual review methods.
2. The proposed system can achieve higher contextual relevance and interpretative accuracy in identifying well control and geological risks compared to conventional text retrieval systems.
3. Insights generated by the LLM assistant will be structured, actionable, and consistent with established operational best practices, thereby positively impacting pre-drill decision-making processes.

Recent advances in LLMs offer a transformative opportunity to unlock the strategic value of these unstructured data sources. Unlike conventional systems, LLMs can comprehend context, synthesize information from multiple documents, and respond to user questions in natural language. When combined with RAG architectures, LLMs can provide geographically targeted, risk-aware recommendations by retrieving and interpreting documentation relevant to a given query—such as nearby well control incidents or pressure anomalies in specific formations. RAG was specifically chosen for this application over traditional fine-tuning approaches because it offers superior flexibility in dynamically incorporating new or updated documents without extensive retraining, thus enabling rapid adaptation to evolving operational data and continuous integration of fresh insights from ongoing drilling activities.

This work presents a framework for an LLM-powered assistant designed to support engineers in the pre-drill planning process. The system enables users to query historical drilling experiences based on coordinates, formations, or complication types, and receive structured, actionable insights derived from offset well data. We simulate the entire pipeline—from document parsing and vector indexing to LLM inference and recommendation generation.

Through this simulation, we evaluate the assistant's performance, response quality, and potential business impact. The results suggest that such a system could help reduce non-productive time (NPT), enhance confidence in mud weight selection, and improve risk awareness in complex geological settings.

Methods

System Architecture

The proposed system utilizes a hybrid architecture, combining a pre-trained LLM with a RAG (Lewis et al., 2020) pipeline. Specifically, the DeepSeek model (DeepSeek-AI, 2024) was selected due to its multilingual capabilities and proficiency in handling extended context sequences of up to 16,000 tokens.

The system architecture comprises four key stages.

Document Preprocessing. Historical drilling reports (daily operational reports in PDF format) are parsed into structured textual segments, such as Daily Operations, Formation Tops, Complications, Mud Properties, Lost Time Details, Foreman Remarks, Personnel on Location, Rig Move Data, Mud Data, and Bottom Hole Assembly (BHA) Information. Regular expressions and domain-specific heuristics are used to extract key metadata, including well identifiers, formation depths, mud weight annotations, cumulative time loss indicators, equipment descriptions, and sensor-based safety parameters. Special attention is given to operational difficulties commonly reported in daily records—such as stuck pipe events, equipment failures, and unplanned interventions, ensuring that the LLM assistant is capable of retrieving and interpreting these high-risk operational scenarios with contextual accuracy.

Embedding and Indexing. These embeddings are stored in a Facebook AI Similarity Search (FAISS) vector database to facilitate fast semantic retrieval. Segments typically contain around 300 tokens with 50-token overlaps; however, larger segments (up to 600 tokens) may be utilized to adequately capture complex operational narratives or detailed incident descriptions, thus preserving contextual coherence.

Query Resolution. Natural-language user queries are standardized using structured templates. The system applies normalization routines and expands domain-specific shorthand expressions to enhance semantic retrieval accuracy. For queries involving location-based risk assessments, the assistant identifies relevant offset wells within a defined spatial proximity using a synthetic coordinate grid. Results are further filtered based on geological formations or operational complication categories, enabling context-aware insights tailored to spatial constraints.

LLM Inference and Generation. Retrieved document segments form structured prompts tailored for the DeepSeek model. The model outputs are standardized into clearly defined structures (incident summary, interval, evidence, recommended actions), ensuring domain consistency and reducing the risk of hallucinations.

To visualize the full workflow, Fig. 1 illustrates the end-to-end architecture — from pdf morning report document ingestion to LLM response generation — including all major processing blocks: parsing, embedding, vector retrieval, query handling, and structured output.

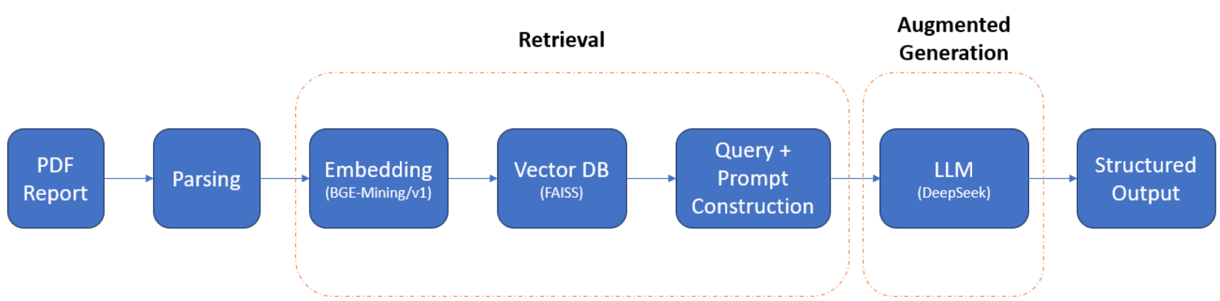

Figure 1—End-to-end architecture of the LLM assistant implementing a RAG pipeline.

Simulation Methodology

A controlled simulation environment was developed to realistically replicate operational scenarios, ensuring the relevance and authenticity of evaluation despite proprietary data constraints. The goal was to rigorously test the system's capability in processing realistic input scenarios, accurately retrieving context-specific information, and generating actionable, high-quality engineering recommendations.

Corpus Construction. Approximately 30 anonymized drilling reports were used, carefully maintaining the authentic narrative structure, operational complexity, and specialized domain terminology common to real-world reports. Reports included detailed descriptions of frequent and critical operational challenges, such as stuck pipe incidents, fishing operations, drill string twist-offs, and abnormal flow events. Each report preserved precise metadata, including mud weight variations, specific depth intervals, cumulative lost time, and comprehensive BHA configurations, closely mirroring actual documentation practices utilized in operational drilling environments. Although coordinate-based filtering was embedded into the retrieval pipeline to enable future location-aware analysis, in this simulation we deliberately selected a set of spatially adjacent wells. As a result, geographic constraints were not enforced during retrieval. Nevertheless, the logic for spatial filtering (e.g., bounding box or nearest neighbor search) is implemented and can be activated as needed in production deployments.

Query Design. An important innovation introduced in this framework is the ability to filter retrieved results based on geographic proximity and geological relevance. For spatially constrained queries, the system resolves input coordinates and identifies nearby wells using a synthetic spatial index. Retrieved documents are then filtered to include only those associated with wells located within a defined radius and matching the same target formation. This functionality ensures that recommendations are not only contextually appropriate but also geographically and geologically specific to the queried drilling location.

To test this capability, a carefully constructed set of five representative queries was developed to simulate realistic pre-drill planning scenarios and typical troubleshooting inquiries encountered by drilling engineers. Queries were phrased in natural language to reflect authentic operational concerns, including:

- "What types of complications occurred after changes in drilling fluid density within nearby wellsin X/Y coordinates?"
- "Summarize historical wellbore instability events in the target formation within a 5 km radius."
- "Highlight any abnormal circulation patterns reported in tight carbonate intervals around the planned trajectory."
- "Retrieve previous operational challenges encountered at similar depths and geological settings near the proposed well."
- "What mitigation actions were applied in wells with recurring tool failures located within the same field block?"

These queries were selected for their diversity in content, complexity, and operational relevance. They rigorously test the system's ability to retrieve accurate historical insights, interpret context, and apply domain-specific reasoning across spatial and geological dimensions.

Manual Evaluation. The outputs generated by the LLM-powered system were assessed by two drilling domain experts using a structured rubric that captured key qualitative dimensions. In addition to evaluating general language generation quality, special attention was paid to the system's ability to satisfy newly introduced spatial and geological filtering constraints.

- Contextual Relevance was judged based on how well the retrieved documentation matched the user query not only semantically, but also in terms of spatial proximity (e.g., wells within 5 km radius) and geological relevance (e.g., same target formation).
- Technical Accuracy was evaluated by verifying the correct classification of operational challenges, such as distinguishing ballooning events from kicks, and ensuring that responses adhered to domain expectations without revealing restricted operational metrics (e.g., depth or pressure values).
- Clarity and Actionability focused on the structure, interpretability, and usefulness of responses for supporting real-world decision-making.

Each response was scored on a 100-point scale, with accompanying expert commentary documenting contextual fidelity, correct application of spatial/geological logic, and compliance with disclosure constraints.

Business Scenario Modeling. A comprehensive business impact scenario was designed to simulate the planning and decision-making process for drilling through a historically problematic carbonate interval with frequent well control incidents. This scenario explicitly incorporated spatial reasoning by defining a set of input X/Y coordinates and target formation characteristics, allowing the system to retrieve only those historical incidents occurring within a 5 km radius and matching the same geological unit.

The simulation compared outcomes between conventional manual review and the LLM-powered assistant equipped with geospatial and formation-based filtering capabilities:

- Analysis Time was evaluated by measuring the duration required to identify and interpret relevant historical incidents.
- Risk Identification Accuracy focused on how effectively each method surfaced subtle yet critical operational signals specific to nearby wells and similar formations.
- Mud Design Confidence assessed how well the assistant supported optimal mud weight selection and mitigation strategy formulation based on geographically and geologically analogous precedents.

This scenario enabled a realistic evaluation of the system's ability to deliver localized and actionable engineering intelligence, demonstrating a clear operational advantage over traditional workflows.

Simulated Results

The controlled simulation conclusively demonstrated significant improvements delivered by the LLM-powered assistant, highlighted through explicit, detailed, and practical examples.

Efficiency Improvement

The traditional manual review process typically required approximately 3–5 hours per query to analyze historical documentation, especially when manually filtering for geographically and geologically relevant precedents. In contrast, the LLM-assisted approach, enhanced with spatial and formation-based filtering, dramatically reduced processing time. Engineers received actionable insights within approximately 15 minutes per query, pre-filtered for wells within a specified radius and formation, eliminating the need to manually sift through unrelated documents.

This efficiency gains not only streamlined the pre-drill planning workflow but also allowed engineers to concentrate on high-impact, contextually relevant information, significantly accelerating the decision-making process while preserving analytical depth.

Accuracy and Relevance

The semantic retrieval precision of relevant document segments reached 92%, enabled in large part by the system's ability to apply geographic and geological filtering during the retrieval phase. This filtering logic helped reduce semantic drift and ensured that retrieved materials reflected true operational analogs in both location and formation.

Domain expert evaluations yielded consistently high average scores of 87/100, reflecting exceptional clarity, technical accuracy, and operational utility of the responses. Experts specifically noted that the system's attention to local geological and spatial context significantly improved the relevance and trustworthiness of the recommendations.

To provide a more granular view of system performance across diverse query types, Table 1 summarizes the evaluation metrics. These include semantic retrieval precision (as measured by expert-defined relevance overlap), expert-assigned response quality scores, and the average response length in tokens. The uniformly high scores across technical scenarios validate the robustness and generalizability of the assistant's architecture in context-specific drilling applications.

Table 1—Evaluation metrics for LLM-generated responses by query type, based on expert assessment and semantic retrieval accuracy.

Query	Retrieval Precision	Expert Score (0–100)	Response Length (tokens)
Stuck Pipe	0.90	89	340
Abnormal Flow	0.92	88	305
Mud Weight Reduction	0.88	85	295
NPT Root Causes	0.91	87	310
Kick Management Practices	0.93	90	330
Bit Damage and Twist-Offs	0.89	86	280
H2S ExposureManagement	0.91	86	290
Formation Tops Prediction	0.87	84	270

Representative Examples

a. **Query:**"Were there any drilling hazards reported near the dolomite interval around coordinates X_1/Y_1?"
   Assistant Response:
   - Retrieved reports from nearby offset wells that penetrated the same dolomite-bearing layer.
   - Identified issues including fluid losses, elevated torque, and reduced rate of penetration.
   - Suggested preventive actions such as loss circulation control and real-time torque monitoring.

b. **Query:**"What measures helped reduce stuck pipe frequency in wells close to X_2/Y_2 in the Lower Jurassic carbonate?"
   Assistant Response:
   - Extracted incident logs from wells within a 5 km radius in the same formation.
   - Highlighted effective measures including adjusted WOB parameters, increased circulation rates, and zonal reaming practices.
   - Recommended incorporating torque/buckling models during planning for high-risk intervals.

c. **Query**: "Summarize wellbore instability challenges reported near the planned trajectory in the target reservoir."
   Assistant Response:
   - Identified consistent problems across spatially adjacent wells with similar geomechanical profiles.
   - Noted recurring instability markers such as high drag, erratic toolface orientation, and pack-off indications.
   - Recommended phased reaming and geomechanical surveillance as mitigative steps.

d. **Query:** "Were there fluid anomalies in nearby wells while drilling through the tight carbonate interval near X_3/Y_3?"
   - Retrieved multiple accounts of unexpected pit volume gains and changes in annular pressure profiles.
   - Flagged ballooning tendencies and early signs of underbalanced flow as potential causes.

   - Suggested real-time flow signature recognition and MW margin adjustments as corrective responses.

e. **Query:**"How were tool reliability issues managed in wells drilled near X_4/Y_4?"
   Assistant Response:
   - Compiled historical equipment failure records from offset wells in the same operational block.
   - Documented common tool integrity problems attributed to high vibrations and thermal cycling.
   - Proposed enhancements including the use of downhole shock absorbers and tool integrity diagnostics between runs.

Business Impact Scenario

A realistic decision-making scenario was constructed to assess the business value of deploying an LLM-powered assistant for planning a well through a historically problematic carbonate interval, known for its frequent well control incidents and geomechanical complications.

This scenario highlighted how the assistant's ability to retrieve spatially and geologically filtered insights from relevant analog wells enabled engineers to make more confident and timely decisions.

- **Without LLM Assistance:**
  - Engineers conducted a manual review of historical drilling documentation, requiring substantial time and effort to identify relevant analogs.
  - Subtle indicators of risk, such as precursors to fluid influx or instability, were often overlooked due to scattered reporting formats and time constraints.
- **With LLM Assistance:**
  - The assistant rapidly retrieved relevant examples from geologically and spatially similar wells using its built-in filtering mechanisms.
  - Historical kick incidents and instability events were surfaced within minutes, with explanations tailored to the target formation and well location.
  - Engineers gained enhanced confidence in mud weight design and risk mitigation strategies, informed by specific analogs and documented field-proven actions.

This scenario confirms that LLM-based systems can significantly reduce planning time, improve risk awareness, and increase operational confidence — especially when applied in complex geological settings where location-specific lessons learned are critical.

Discussion

The simulation results reinforce the transformative potential of integrating LLMs with RAG pipelines in upstream drilling operations. By leveraging advanced language models, drilling engineers gain unprecedented efficiency in accessing, interpreting, and utilizing historical drilling reports. The assistant's ability to convert unstructured documentation into structured, geologically and geographically relevant insights enables more precise, context-aware decision-making.

Compared to traditional manual analysis, which often requires hours of work and extensive domain knowledge, the LLM-based assistant produced contextually accurate, technically sound, and actionable recommendations in a fraction of the time. This efficiency gain is particularly valuable in complex carbonate formations, where historical risks—such as stuck pipe incidents, well control events, and fluid anomalies—can have significant operational impact.

Moreover, the assistant's capacity to resolve spatially constrained queries (e.g., filtering results from nearby wells in the same formation) enhances its applicability in practical field planning. The examples presented in the simulation demonstrate how such spatial targeting supports localized risk identification and improves mud design confidence, based on actual field analogs.

Practical Implementation and Scalability Considerations

The presented solution is designed with scalability and deployment readiness in mind. Its architecture allows for real-time implementation in operational environments, assuming the availability of appropriate infrastructure. This typically involves GPU-accelerated computing to support inference speed and memory demands. Depending on regulatory and security constraints, the system can be deployed either on secure on-premise servers or through hybrid cloud infrastructures that balance accessibility with compliance.

Operational robustness requires the implementation of additional safeguards. Confidence scoring should be introduced to estimate the reliability of each generated response. Post-processing validation and consistency checks can help reduce the risk of hallucinated or misleading outputs. These elements contribute to trustworthiness in high-stakes engineering contexts.

Beyond unstructured document analysis, integration with structured data sources can further enhance system capabilities. Incorporating real-time sensor feeds or predictive models for formation pressure allows the assistant to support more comprehensive, multi-source decision-making workflows. Such hybridization strengthens situational awareness and provides engineers with deeper contextual insights.

To ensure continuous improvement and adaptability, the system must include mechanisms for user feedback collection and active learning. Regular fine-tuning based on user interactions will help align the assistant's responses with field-specific expectations, maintain technical accuracy over time, and expand its utility in new drilling domains or formations.

Study Limitations and Future Directions

While the simulation environment was carefully constructed to emulate realistic drilling scenarios, it remains constrained by the use of a limited set of anonymized reports. These documents, though representative in structure and content, do not fully capture the diversity and complexity of actual field reports, which may include graphical content, handwritten annotations, and variable formatting standards. Future development should address these aspects through the application of advanced OCR pipelines, multimodal pretraining strategies, and expanded corpora drawn from a wider range of geological contexts.

In terms of architecture, the current study relies on the original RAG framework (Lewis et al., 2020). To enhance robustness and response accuracy, future implementations may adopt advanced retrieval-generation hybrids such as Self-RAG (Asai et al., 2023), which introduces self-reflective retrieval steps, or RAG-Robust (Xiang et al., 2024), which improves factual consistency in noisy environments. Other promising directions include Retriever-Ranker systems (Lu et al., 2022), which employ dynamic reranking and iterative refinement to improve the precision and clarity of outputs.

These architectural improvements will be especially valuable in complex use cases that require nuanced understanding and high interpretive accuracy. For example, they could support disambiguation between incident types such as kicks and ballooning, facilitate adaptive retrieval across multiple wells or stratigraphic intervals, and enable confidence-based response filtering in risk-sensitive decision contexts.

More broadly, these developments lay the groundwork for agentic LLM systems, where dedicated components specialize in different tasks across the information pipeline—from retrieval and validation to recommendation—fostering greater transparency, modularity, and explainability in AI-assisted operational workflows.

Conclusions

This study confirms the strategic value of integrating large language models with retrieval-augmented generation frameworks to extract actionable insights from historical drilling documentation. Through simulation on a realistic and curated corpus, the LLM-powered assistant demonstrated the ability to deliver responses that are not only technically accurate and operationally relevant but also spatially and geologically specific to the queried drilling location.

A key innovation of the proposed framework lies in its ability to interpret user queries with embedded location context. By resolving input coordinates and filtering retrieved content based on geographic proximity and target formation, the system ensures that its recommendations are grounded in highly relevant historical analogs. This coordinate-aware retrieval mechanism adds a new dimension to risk assessment workflows, enabling engineers to evaluate prior incidents, complications, and mitigation strategies with greater spatial precision.

The assistant significantly reduced analysis time, enabling engineers to shift from hours-long manual review to delivering insights within minutes. It achieved a semantic retrieval precision of 92 percent, with domain experts assigning average relevance and accuracy scores of 87 out of 100. These results reflect the system's strong performance in identifying and distinguishing complex operational anomalies such as kicks, stuck pipe, ballooning, and abnormal flows.

Practically, the solution enhances pre-drill planning by embedding historical knowledge into current decision workflows. This enables better anticipation of risks and improved design of drilling programs, especially in geologically complex or high-uncertainty areas.

Looking ahead, the system can be further strengthened by incorporating real-time sensor data, expanding to multimodal inputs, and refining its retrieval strategy through adaptive learning based on user feedback. These improvements would support the transition from retrospective analysis to real-time, context-aware decision-making.

In summary, this research highlights the immediate applicability and future potential of LLM-based systems in upstream operations. By combining technical insight with spatial intelligence, the assistant provides a foundation for smarter, safer, and more informed drilling practices.

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